Algorithm 6 Evolved selection: suppress clones and explore quality-gated niches

Input Population P; tournament size k; geometric parameter p; maximum complexity L; global normalized complexity frequencies f_c; parsimony scale λ.

$S \leftarrow$ uniformly sample $\min(k, |P|)$ members without replacement
for $i \in S$ **do**
  $c_i \leftarrow \textsc{Complexity}(i)$; $\ell_i \leftarrow \textsc{Loss}(i)$
  $s_i \leftarrow \textsc{Cost}(i) \exp(\lambda f_{c_i})$ if frequency weighting is enabled,
    otherwise $s_i \leftarrow \textsc{Cost}(i)$; use $f_c = 0$ outside $1 \le c \le L$
$\pi \leftarrow \textsc{Argsort}(s)$; $\widetilde{s} \leftarrow s$
for $a = 2, \ldots, |S|$ **do**
  $i \leftarrow \pi_a$
  if some $j \in \{\pi_1, \ldots, \pi_{a-1}\}$ satisfies $c_i = c_j$ and $|\ell_i - \ell_j| \le 10^{-6} \max(|\ell_i|, |\ell_j|, \epsilon)$ **then**
    $$\widetilde{s}_i \leftarrow \begin{cases} 1.35 s_i, & s_i > 0, \\ s_i / 1.35, & s_i < 0, \\ 0, & s_i = 0 \end{cases}$$ # One sign-safe penalty per clone
$\rho \leftarrow \textsc{Argsort}(\widetilde{s})$
if $\mathrm{Unif}(0,1) < 0.10$ **then**
  $H \leftarrow$ first member of each valid complexity in ρ order
  if $H \neq \varnothing$ **then**
    $H_q \leftarrow$ first $\lceil |H|/2 \rceil$ members of H # Keep the better half
    return $\arg\min_{i \in H_q} (f_{c_i}, \widetilde{s}_i)$ # Rarest niche; quality breaks ties
Draw rank $r \in \{1, \ldots, |S|\}$ with $\Pr(r) \propto p(1-p)^{r-1}$
return S_{ρ_r} # For $p = 1$, choose the best adjusted score

Clone detection uses complexity and nearly equal loss as a phenotype proxy; it does not compare expression strings. All earlier members in the *original* ranking can establish a clone match, including earlier clones. Niche rarity uses global search frequencies, even when frequency weighting of tournament costs is disabled. If no valid niche exists, selection falls back to the geometric tournament.